

TY: Facilitating social engagement to mitigate loneliness during transitory periods

Melissa, M.F., Li

Faculty of Arts and Science, McGill University
Melissa.li2@mail.mcgill.ca

Bronwen, W., Lathrop

Faculty of Arts and Science, McGill University
Bronwen.lathrop@mail.mcgill.ca

ABSTRACT

The state of loneliness in modern society has been referred to by researchers as an epidemic, the rate of which has been exacerbated by COVID-19. Loneliness can have significantly negative impacts on human wellbeing. In this paper, we propose gratitude interventions as a potential solution to improve prosocial behaviour, social connectedness and to function as a mediator of the negative affect associated with loneliness. We present a multi-modal gratitude intervention journaling app that leverages research in social isolation and gratitude intervention and aims to alleviate some of the consequences of loneliness.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

loneliness, gratitude journaling, transitory periods

ACM Reference Format:

Melissa, M.F., Li and Bronwen, W., Lathrop. 2022. TY: Facilitating social engagement to mitigate loneliness during transitory periods. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3491101.3514494>

1 INTRODUCTION

Loneliness is a human emotion characterized by an emotionally painful experience that transpires when an individual assesses their interpersonal relationships to be fewer and or lesser in quality than desired. The state of loneliness in modern society has been referred to by researchers as an epidemic [8]. A 2021 national survey of American adults from Harvard University researchers reported that 36% of respondents described feeling lonely “frequently” or “almost all the time or all the time” in the four weeks prior to the survey — a number that includes more than 60% of young people ages 18-25 [21]. The rate of loneliness has been exacerbated by the COVID-19 global pandemic, with close to fifty percent of young adults reporting increased feelings of loneliness since the pandemic began [21]. The impact of loneliness on mental health cannot be overstated; it has been shown to increase depressive symptoms, perceived stress,

anxiety, anger, as well as diminish optimism and self-esteem [7]. Moreover, loneliness has been associated with personality disorders, psychoses, suicide, cognitive decline over time and even increased risk of Alzheimer’s Disease [7]. Neuropsychological research has indicated that the impact of loneliness can be observed in the brain; greater loneliness is associated with less activation in the medial prefrontal cortex — a region of the brain involved in mapping social ties and closeness [5]. However, the damaging effect of loneliness is not limited to mental health and cognitive functioning. A 2020 study from the National Academies of Sciences, Engineering, and Medicine found that social isolation significantly increases the risk of premature death. In short, the impact of loneliness on mental and physical well-being is extensive and punishing.

Loneliness does not discriminate amongst populations. Historically, psychology research has focused on the chronic aspect of loneliness which has led to an abundance of research on older adult populations [10], as they are at increased risk of experiencing continued loneliness and social isolation. This is because they are more likely to encounter challenges like living alone, loss of loved ones, chronic illness, and decline in communication skills [14]. However, other age groups from young adults to people at midlife also experience loneliness, typically spurred from life events that result in separation from their families, community networks and friends during major life changes. During such stages, individuals are vulnerable to episodic periods of loneliness. A 2020 US Report found that individuals who had transitioned to a new job in the past six months had loneliness scores more than six points higher than those who had held their positions for over a decade [4]. The same report found that the developmental stage at which people tend to make major transitions like starting university and leaving home [17]— between 18 and 22 — had the highest average loneliness score relative to any age group [4]. While some groups are at higher risk, it’s apparent that people of all ages and backgrounds experience loneliness. Accordingly, it is fitting that instead of focusing on the chronic aspect of loneliness, we should instead shift our focus on the common denominator between age groups: how loneliness transpires from transitory periods. According to Lisa Thomas and colleagues, transition is both the process of change over time and the resources required to engage with that change. This definition calls to attention the possibility of experiencing a transitory period without the capability to adequately navigate it [8].

We designed TY, a multi-modal journaling app that aims to help individuals who are experiencing any life event where there is change that sparks sentiments of loneliness. TY leverages gratitude expression and journaling as means to facilitate pro-social behavior and social intentionality in hopes of promoting strengthening of social bonds and reducing feelings of loneliness (Figure 1).

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3514494>

2 RELATED WORK

Intervention strategies such as improving social skills and enhancing social support have been identified among effective loneliness reduction interventions [13]. Gratitude can be described as a positive emotional response characterized by appreciation, thankfulness, and a sense of wonder [19] that is robustly associated with features of high-quality, healthy social relationships [15]. We have identified gratitude as a practice with the potential to combat loneliness because of this intrinsically social-relational element of gratitude [15] and research that has demonstrated its positive impact on well-being. Such research has shown that gratitude increases likelihood of prosocial behavior, trust in others, and helps to build and reinforce social bonds [9]. Notably, prosocial behavior has been negatively associated with loneliness and positively associated with peer acceptance at all ages [18]. Interestingly, gratitude has been associated with increased activity in the medial prefrontal cortex, the same region previously noted to activate less with increased loneliness [5]. One method that has been shown to cultivate gratitude is the practice of gratitude writing [9]. This practice generally involves several sessions during which participants journal or write letters to express gratitude based on guiding prompts [22, 23]. Research suggests that gratitude writing facilitates feelings and expressions of gratitude and associated benefits [15, 19, 20]. In their 2012 study, Toepfer et al. found that depressive symptoms—symptoms that are positively associated with loneliness [3]—were decreased following gratitude writing activities. Gratitude writing may be especially beneficial for individuals suffering from loneliness because of the focus attention on positive interpersonal connections [5].

3 RESEARCH

In recent years there has been a surge in availability of mindfulness, note-taking, gratitude, and habit tracking apps, however, there is not a single one that specifically caters towards gratitude journaling for social encouragement. Therefore, our research process was initialized by familiarizing ourselves with available apps across domains and conducting semi-structured interviews for a deeper understanding of the intersection that we were interested in.

Sixteen participants between the ages of 18 and 78 were recruited across four demographics: young adults (18-25 years), adults (25-39 years), midlife (40-64 years) and older adult (over 65 years). All individuals recruited expressed previous experiences of mild to severe loneliness and were currently going through a transitory life period which includes, but is not limited to: completing their degree, ending a romantic relationship, starting a new job, moving cities and taking on new caretaker role by having an older-adult parent move-in.

Our goal was to understand how expression of gratitude mediated by technology could help counteract the cyclical nature of loneliness during transitory periods. While it was considered, we ultimately decided not to include a diary study during this iteration. Based on this, we were careful to pose questions in our semi-structured interviews that could provide us with a sufficient foundation for our design rationale for a preliminary solution to the problem. We asked a range of broad to specific questions allowing the participants to comfortably share what they deemed relevant. Although participants answers varied from our posed questions,

in many instances we were able to find underlying reasoning that clearly led us in the right direction for our design ideation. For example, when we first asked participants about preferred methods or mediums for expression, a variety of answers received included blank documents, notebooks, voice memos or even scrap paper. Upon further discussion we found that the common reasoning for almost all participants was that little to no distraction was preferred when expressing. As we later discuss in detail, this inspired our minimalistic design for the expression screens (Figure 2).

During these interviews we were also able to strengthen previous notions we had about journaling, such as the fact that it was a highly intimate and personal process, while also learning from unanticipated responses. For example, curious as to if and how we should design notifications, our discussions lead to the discovery that majority of the participants were inclined to journal when “inspiration strikes” rather on an intentional interval. When asked about scheduling time to write, it was almost unanimously agreed upon that it turns the process into a chore, rather than the cathartic act they enjoy. This finding in conjunction with literature review about continued expression directly inspired the way we engaged with users in content design. Moreover, about half of the participants usually had prompts they used to get them started and those who did not reacted positively to the idea, sharing that prompts would likely make sustained efforts in expression more likely. While this finding did not necessarily equate to a completely new feature, it supported our rationale for including prompts on the hyper-minimalistic expression screens. In another exploration, we asked participants if they often re-read what they had previously written. Of those who did, reasons were primarily sentimental in nature as seeing the change or growth over time provided positive emotions. This effected the way we prioritized and designed the entries screen (Figure 3). Overall, we heavily relied on detailed semi-structured interviews to ensure that our design requirements were rooted in real user stories and that we had sufficient insight to create a viable design solution.

4 DESIGN REQUIREMENTS

Based on both our literature reviews and semi-structured interviews, we began to derive design requirements with the supplemental design research methods such as affinity diagramming, journey mapping, persona building and rapid prototyping. Each of these requirements were further grounded by secondary research.

4.1 Encourages Sustained Use

Research has shown that actions like gratitude writing are most effective when participants have effortfully endorsed and committed to the intervention on a continued basis [11]. However, people tend to adapt and become desensitized to positive events when there is continued focus on them. In fact, one study by Lyubomirsky et al. suggested that people who wrote in gratitude journals once a week for six weeks experienced the positive benefits associated with gratitude writing whereas participants writing three times a week did not [12]. Putting these findings into action, our design needs to encourage sustained and effortful gratitude in order to help the user achieve an ideal level and timing of gratitude exercises to make a sustained positive impact based on real research. Keeping

the purpose of gratitude interventions at the forefront of our design will help users to remain committed to the benefits of gratitude writing.

4.2 Supports Prosocial Behavior while Remaining Non-Invasive

The find-and-bind theory of gratitude, first discussed by Algoe et al. in 2008, suggests that gratitude is a positive emotion that serves an evolutionary purpose. According to the find-and-bind theory, gratitude strengthens one's relationship with a responsive interaction partner [1]. The find-and-bind theory is characterized by the concept that individuals who feel grateful perceive their relationship with their benefactor as closer and behave in a manner consistent with communal relationship norms. That is, post-gratitude social behaviors are marked by relationship strengthening rather than simply repayment [1, 2]. There is also evidence to suggest that people who are thanked for their actions are more likely to behave in a prosocial manner towards the expressor and to others in general moving forward [1, 6]. In short, expressions of gratitude have been shown to facilitate initiation and maintenance of prosocial behavior [6] as well as increase trust in others and build social bonds [2, 9].

Some literature suggests that loneliness itself is a source of shame [16]. This might include shame associated with feelings of inadequacy and guilt at the inability to form fulfilling or meaningful interpersonal connections. Some studies even express concerns about underreporting loneliness due to the stigma associated with being identified as lonely [16]. In accordance with this research and insights from our interviews on the vulnerable nature of journaling, we aim to make our designs non-invasive for the user and privacy focused.

4.3 Simple and Accessible

Our last design requirement is typically an unstated expectation of designs, but we felt that it was appropriate to stipulate this explicitly so that we prioritize this. Designing for virtually all age groups with a common interest can be challenging, but based on the overt as well as covert needs identified during our semi-structured interviews, and supported literature review, keeping the application simple not only provides accessibility but aligns with the goals of the gratitude expression process — to focus on what's most important and drown out the noise.

5 DESIGN OF TY

Based on our research and design requirements gathered, we began to design our app. We did this through sketching, wireframing and low-fidelity prototyping. Throughout this process we paid close attention to the user experience, but equally importantly, the design of the user interface itself, and the role it would play in supporting the desired outcome for our users. The overall interface is two dimensional, with an off-white background and clean black lines for separation inspired by notebook pages, a compromise for our non-digital journaling enthusiasts, and a clean feel for others. All buttons have clearly defined boundaries and follow our design system which includes primary, active and normal states. Pops of muted color were used on most screens to convey a positive but subdued look, while the actual screen for journal entries is left plain,

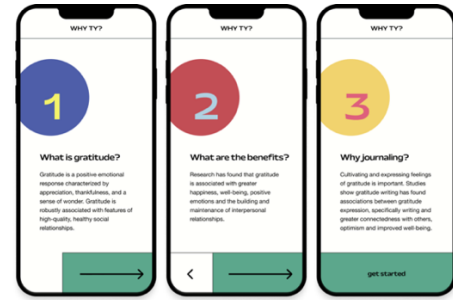


Figure 1: Onboarding screens explaining gratitude, the benefits, and the role of journaling

with hopes to reduce distraction for the user — a common theme brought up during interviews. Notably, we chose warmer hues due to the distortion of perception of cool tones with aging vision. Iconography was also kept relatively straight-forward. The user experience as seen through feature selection and design include (1) focus on user inclusion, (2) multi-modal expression, (3) multi-surface notifications, (4) intelligence guided expression and lastly, (5) sentiment analysis for closed loop design.

5.1 User Inclusion into the Process

During the onboarding experience, we define gratitude, the benefits of doing practicing gratitude and how journaling can be an effective method to do so (figure 1). According to Lyubomirsky et al., expressing gratitude and optimism did not generally increase well-being unless a person was truly cognizant of the exercises' purpose and motivated to improve their own happiness [11]. Therefore, the decision to include this as a primary feature directly supports our design requirement of endorsement of sustained usage by ensuring users are optimally set up understand, endorse and commit to the intervention.

5.2 Multi-Modal Input for Creative Expression

Based on our prioritization of accessibility, the variety of mediums and methods participants mentioned using to express themselves, and the general time-consuming nature of text-based journaling on certain devices such as smartphones, we designed the journaling experience to be multi-modal. Users have the liberty to choose between typed text, speech-to-text or even draw/doodle (Figure 2). This decision was supported by heuristic evaluation of other note-taking apps. While they are not specifically designed to support gratitude journaling, we wanted to ensure our design does not disadvantage the user in any way. Instead, this design helps make gratitude expression using TY more flexible, less formal, and equally robust as other note-taking alternatives. Additionally, based on interview insights and research on sustained use, TY sends gentle prompts to get users started in their entries if they wish. They are seen in a lighter color to easily fade out of focus if unwanted.

5.3 Multi-Surface Capabilities

While smartphones are the most common surface used amongst our participants, tablets have been noted to provide added benefits

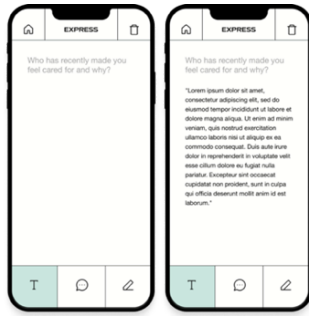


Figure 2: Expression screens where users can type, speak or draw to express gratitude

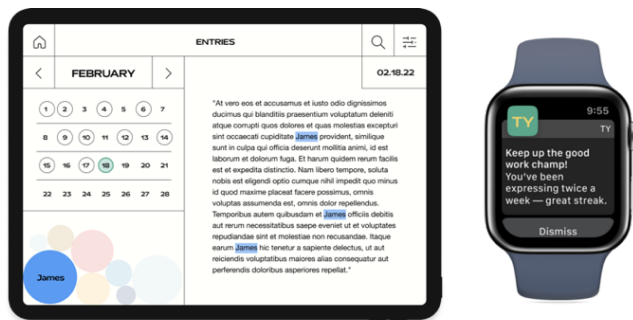


Figure 3: Other surfaces TY is available on. Tablet (left) depicts entries screen and smart-watch (right) displays push notification

for a variety of sub-populations. Generally, tablets are capable of added accessibility, as the larger screen size provides flexibility to increasingly design with legibility and motor dexterity in mind. Furthermore, our multi-surface design is doubly useful when considering the multitude of preferred methods of expression — today's tablets are optimized for longform writing, drawing, and much more as seen by popular apps such as Notability and Procreate. In addition to the tablet, we were inspired to design for a companion smart watch app that helps positively reinforce gratitude expression and usage of TY, with the ultimate goal of promoting prosocial behavior as outlined in our requirements (Figure 3).

5.4 Sentiment Analysis for Privacy Focused Social Suggestions

Lastly, based on evidence that supports self-reflection when journaling, benefits of self-awareness into the practice, and our design requirement of supporting prosocial behavior, we designed for the insight tool (Figure 4) to analyze user entries to help identify common names that are frequently brought up in their entries. Notably, based on shared opinions of intimacy and privacy, these names are not connected to any particular social profiles or external accounts. Names identified are simply from sentiment analysis alone, and there is no third-party app integration in this design. Users can choose the timescale of the entries they want to analyze and



Figure 4: Insight screens provide users a glance at frequently mentioned names (left) and their most commonly associated themes (right)

names will be brought up accordingly. Sizes of circles represent the frequency which those names are mentioned. Furthermore, users can explore themes, topics and words commonly used in entries pertaining to individual people. Our hope is that TY helps guide users to identify those who they may want to express feelings of gratitude towards, subconsciously want to connect with or generate potential interest or commonalities that can spur conversations that create stronger social bonds. The design was also meant for users to visualize their community and decide where they might want to be more socially intentional so they can form new social bonds or strengthen old ones, thus improving feelings of loneliness and desire for connection. While studies have shown benefits to reflective and expressive gratitude writing, interpersonal gratitude oriented towards others has been shown to be particularly effective in enhancing emotional well-being when outwardly expressed [15].

6 USER SCENARIO

There is a plethora of transitory life periods that spur loneliness where TY could provide reasonable support. To illustrate we created a user scenario for Charlotte, a woman who has recently relocated to a big city for an exciting new job. Since moving, she has been feeling very homesick. Charlotte misses her friends and family and is having a hard time forming meaningful relationships in this new city. She spends her time at home and feels awkward and uncomfortable at work, where she mostly keeps to herself. Charlotte keeps to herself during work hours. Charlotte has been making a conscious effort to improve her feelings of loneliness and homesickness. She gets prompts from her TY smartphone app that encourages her to make gratitude writing entries. After a few weeks, TY notifies Charlotte that she has expressed gratitude towards a friendly coworker of hers, Annie, several times and encourages her to let Annie know. Charlotte is redirected to her messenger app where she sends Annie a text message thanking her for her kindness during these difficult times. Annie and Charlotte begin a conversation that leads to a budding friendship.

7 DISCUSSION AND CONCLUSION

The design that has been outlined in this paper presents a possible solution to support individuals struggling with the experience of loneliness by way of leveraging the benefits of gratitude writing and expression. TY is a multi-modal journaling interphase that has been developed with the intention of facilitating social connectedness through gratitude writing and expression. Keeping research at the forefront of our design process, TY has been developed to encourage sustained and effortful gratitude writing with prompts to users consisting of gentle reminders to complete weekly gratitude writing goals and reminders of when weekly goals have been met. We have also centred our design around fostering the inherently social aspects of gratitude writing into outward expressions through sentiment analysis and recommendation features. Our hope is that TY helps users to identify people in their lives they are especially grateful for and let them know in order to strengthen social connectedness and overall well-being. If we had more time, we would have ideally included a diary study with a few of the participants we interviewed in our research phase to better understand the nuances of gratitude journaling and its immediate impacts on users lives. Looking forward, our next steps include evaluation of our app by users across age groups, ability and life experience in order to further improve interface accessibility. Recent research has suggested that while gratitude interventions have some positive impacts, previous works may have over-emphasized the associated benefits in some populations [11]. To that end, we aim to collect a larger sample of information concerning the long-term effects of gratitude interventions like TY in individuals experiencing loneliness.

We believe that while rooted in real life user stories and experiences, our design solution has only begun to scratch the surface of how a gratitude expression system such as TY can help social engagement and aid alleviation or complete mitigation of loneliness for those experiencing transitory periods, and are excited by potential future iterations of this work.

REFERENCES

- [1] Holt-Lunstad, J. The Potential Public Health Relevance of Social Isolation and Loneliness: Prevalence, Epidemiology, and Risk Factors. *Public Policy & Aging Report*, 27, 4 (2018), 127-130.
- [2] Weissbourd, R. M., M.; Lovison, V; and Torres, E. *Loneliness in America: How the Pandemic Has Deepened an Epidemic of Loneliness and What We Can Do About It* Harvard Graduate School of Education, 2021.
- [3] Hawkey, L. C. and Cacioppo, J. T. Loneliness matters: a theoretical and empirical review of consequences and mechanisms. *Ann Behav Med*, 40, 2 (Oct 2010), 218-227.
- [4] Courtney, A. L. and Meyer, M. L. Self-Other Representation in the Social Brain Reflects Social Connection. *J Neurosci*, 40, 29 (Jul 15 2020), 5616-5627.
- [5] National Academies of Sciences, E. and Medicine *Social Isolation and Loneliness in Older Adults: Opportunities for the Health Care System*. The National Academies Press, Washington, DC, 2020.
- [6] Cigna. *Loneliness and the Workplace: 2020 U.S. Report*. 2020.
- [7] Sandre, A., Bagot, R. C. and Weinberg, A. Blunted neural response to appetitive images prospectively predicts symptoms of depression, and not anxiety, during the transition to university. *Biol Psychol*, 145 (Jul 2019), 31-41.
- [8] Thomas, L., Orme, E. and Kerrigan, F. Student Loneliness: The Role of Social Media Through Life Transitions. *Computers & Education*, 146 (2020/03/01/ 2020), 103754.
- [9] Masi, C. M., Chen, H. Y., Hawkey, L. C. and Cacioppo, J. T. A meta-analysis of interventions to reduce loneliness. *Pers Soc Psychol Rev*, 15, 3 (Aug 2011), 219-266.
- [10] Toepfer, S. M., Cichy, K. and Peters, P. Letters of Gratitude: Further Evidence for Author Benefits. *Journal of Happiness Studies*, 13, 1 (2012/03/01 2012), 187-201.
- [11] O'Connell, B. H., O'Shea, D. and Gallagher, S. Feeling Thanks and Saying Thanks: A Randomized Controlled Trial Examining If and How Socially Oriented Gratitude Journals Work. *J Clin Psychol*, 73, 10 (Oct 2017), 1280-1300.
- [12] Kerr, S., O'Donovan, A. and Pepping, C. Can Gratitude and Kindness Interventions Enhance Well-being in a Clinical Sample? *Journal of Happiness Studies* (01/01 2014).
- [13] Sletta, O., Valås, H., Skaalvik, E. and Sobstad, F. Peer relations, loneliness, and self-perceptions in school-aged children. *Br J Educ Psychol*, 66 (Pt 4) (Dec 1996), 431-445.
- [14] Wong, Y. J., Owen, J., Gabana, N. T., Brown, J. W., McInnis, S., Toth, P. and Gilman, L. Does gratitude writing improve the mental health of psychotherapy clients? Evidence from a randomized controlled trial. *Psychother Res*, 28, 2 (Mar 2018), 192-202.
- [15] Witvliet, C. v., Richie, F. J., Root Luna, L. M. and Van Tongeren, D. R. Gratitude predicts hope and happiness: A two-study assessment of traits and states. *The Journal of Positive Psychology*, 14, 3 (2019/05/04 2019), 271-282.
- [16] Toepfer, S. M. and Walker, K. K. Letters of Gratitude: Improving Well-Being through Expressive Writing. *The Journal of Writing Research*, 1 (2009), 181-198.
- [17] Campagne, D. M. Stress and perceived social isolation (loneliness). *Archives of Gerontology and Geriatrics*, 82 (2019/05/01/ 2019), 192-199.
- [18] Lyubomirsky, S., Dickerhoof, R., Boehm, J. K. and Sheldon, K. M. Becoming happier takes both a will and a proper way: an experimental longitudinal intervention to boost well-being. *Emotion*, 11, 2 (Apr 2011), 391-402.
- [19] Lyubomirsky, S., Sheldon, K. M. and Schkade, D. Pursuing Happiness: The Architecture of Sustainable Change. *Review of General Psychology*, 9, 2 (2005/06/01 2005), 111-131.
- [20] Algoe, S. B. Find, Remind, and Bind: The Functions of Gratitude in Everyday Relationships. *Social and Personality Psychology Compass*, 6 (06/01 2012).
- [21] Bartlett, M. Y., Condon, P., Cruz, J., Baumann, J. and Desteno, D. Gratitude: prompting behaviours that build relationships. *Cogn Emot*, 26, 1 (2012), 2-13.
- [22] Grant, A. and Gino, F. A Little Thanks Goes a Long Way: Explaining Why Gratitude Expressions Motivate Prosocial Behavior. *Journal of personality and social psychology*, 98 (06/01 2010), 946-955.
- [23] Ong, A. D., Uchino, B. N. and Wethington, E. Loneliness and Health in Older Adults: A Mini-Review and Synthesis. *Gerontology*, 62, 4 (2016), 443-449.